

URBANIZATION TRAJECTORIES IN NIGERIA: COMPARATIVE ANALYSIS USING CENSUS DATA

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INTRODUCTION:

Urbanization in west Africa is happens to have the most rapid rate worldwide, with the entire urban population growing at about 3.5% every year majorly higher than the global average 1.8% (UN, 2024). The increase in urbanization is mainly sourced by rural-urban migration, natural population growth as well as designating some rural area to urban center for various reasons. In West Africa, countries like Nigeria, Ghana, and Senegal shows different urbanization experience I their respective regions which has shaped their economic structures, the governing system and various policies targeted at development.

Nigeria being the most populated countries in West Africa and Africa at large has experience magnificent urban growth in Lagos, Abuja and Port Harcourt (Bloch, R., Monroy, J., & Ojo, A., 2015). Reasons for different urbanization path is an important aspect which helps policy-maker, urban and regional developers. Urban growth has been instrumental to opportunities such as economic growth, innovation, improved service and as well possess challenges like infrastructure overuse, unemployment and under-employment as well as environmental degradation (Henderson, 2003), (Glaeser, 2011). This article seeks to addresses the quantitative analyze and compare urbanization in Nigeria over the selected period of time using census data, with focusing on demographic, economics and spatial aspect of the urban growth.

Rationale and Significance

In light of the importance of urbanization to the development of West Africa countries, it has been observed from studies focused on singular countries or qualitative assessments (Parnell, 2011). There is a comparative quantitative gap that has methodically examine urbanization trajectories across different countries using a standard metric and statistical tools. This article si aimed to;

1. State empirical facts on urbanization growth rate, patterns and drivers in Nigeria
2. Proffer good practices and policies from countries with a more suitable and sustainable urbanization patterns.
3. Educate regional and urban planning as well as development various strategies for Nigeria and other West African countries
4. Use python-based analytical method to replicate similar result for other region in West Africa regions

LITERATURE REVIEW

THEORETICAL FRAMEWORK

Theory of Demographic Transition

Thompson (1929) and Notestein (1945) were of the opinion that the society at large is at the state of transitioning from high natural population growth in the rural area to low natural population growth with urbanization as a deciding factor in this transition, as population concentrate in cities with modernization.

Harris-Todaro Model

According to (Harris, J.R., & Todaro, M.P, 1970) examined rural-urban migration on the basis of expected income differences. This framework is focused on applying to understand urbanization drivers, as migration being the main source of urban growth in West Africa (Fox, S., & Sekkel, T., 2019).

Urban Bias Theory

Lipton (1977) was of the opinion that systemization of policies by government favors urban area in terms of resources sharing, creating incentives for rural-urban migration and rapid urbanization. This framework better explains urbanization that is policy driven that is observed across developing nations.

Urbanization-development Nexus

Henderson, (2003) and Glaeser,(2011) made a suggestion that a positive interaction between urbanization and economic development, using cities to serve as engines for growth through agglomeration economies, innovation and productivity gains.

Empirical Studies on Nigeria

Potts, 2012 recorded that in Africa urbanization differs from historical western patterns, characterized by “urbanization without growth” with cities expanding regardless of limited industrialization. West Africa urban population will likely double by 2050 creating unprecedented challenges on infrastructure and services (UN-Habitat, 2014). Fox, S. et al (2019), found that West Africa’s Urbanization is driven more by migration compared to natural increase contrasting with eastern africa where natural increase is a major role. Nigeria urbanization and housing challenges (Jiboye, 2011)is experiencing a geometric growth that is outpacing infrastructural development which is causes concern in the increasing rate of informal settlements. Examining Lagos’s population dynamics, it was estimated that about 85 to 90 percent of the populace per hectare in some part of the state, hence making it highest densely populated area in africa (Bloch, et al 2015). (Gandy, 2006) also described Lagos as a “megacity” with it entire population to be about 15 to 21 million, forcing a huge impact of data collection challenges and challenges of capturing informal settlement populations

Research gap

There is limited comparative quantitative research using standardized census data and statistical methods to analyze urbanization. This study fills that gap by;

1. Use consistent metric (urbanization rate, primacy, urban growth rate) in Nigeria from 2000 to 2020
2. Use Python-based statistical analysis for transparency and replicability
3. Examine temporal trends to capture recent changes
4. Link urbanization patterns to demographic and economic indicators

Research Questions

What are the comparative urbanization trajectories in Nigeria from 2000 to 2020 and what demographic and economic factors explain the difference?

1. What are the urbanization rate and temporal trends in Nigeria between 2000 to 2020
2. How do urban primacy level, that is concentration in large city, differ one from another and how they evolved over time?
3. What are the annual urban population growth rate and how do they compare to national population growth rates?
4. What are the spatial distribution patterns of urban cities (mega-cities, secondary cities, small towns)?
5. How do demographic factors like fertility and mortality rate and age structure correlate with urbanization patterns?

Research Objectives

In order to achieve the stated research question the following objectives must be considered in getting a comparative quantitative analysis of urbanization trajectories in Nigeria over the span of 2000 to 2020

1. To calculate urbanization rate in Nigeria for the period of 2000 to 2020
2. To calculate and analyze urban primacy indices (share of urban population in the large cities).
3. To determine annual urban population growth rates and compare them with overall national population growth rate.
4. To visualize the spatial distribution of urban centers and assess urban system hierarchies (primary vs balanced distribution).
5. To analyze the associations between demographic variables like total fertility rate, life expectancy, median age and urbanization patterns across major cities in Nigeria

Methodology

This study uses a comparative quantitative research design using secondary data analysis. The approach is

- Cross-sectional and longitudinal: Comparing four major states such as Lagos, Oyo state, Abuja, and Port Harcourt at multiple points (2000, 2005, 2010, 2015, 2020)
- Descriptive and analytical: Calculating urbanization metrics and testing relationships with explanatory variables
- Python-based: All data processing and analysis conducted using python for transparency and replicable.

Study State in Nigeria Selection

Cities that were selected based on

1. Population density
2. Varying economic development level
3. Data availability and quality

Total Population	Urban Population	Rural Population	Major City	State	Urbanization Rate	Pop Growth Rate	Urban Growth Rate
14,368,332	12,966,892	1,401,440	Lagos Metropolis	Lagos	90	14	17
7,840,864	4,154,449	3,686,415	Ibadan	Oyo	53	11	18
7,303,700	5,186,664	2,117,036	Port Harcourt	Rivers	71	11	16
4,370,000	4,197,300	172,700	Abuja	FCT	96	33	35

Source: United Nations (2024); World Bank (2024)

Data sources

Main data sources:

1. United nations World Urbanization prospects (2024 Revision)
2. World Bank World development Indicator
3. National Census Data
4. UN Population Division Demographic Indicators

Data Access:

- All data sources are publicly available and free to access
- World Bank data accessible via python (`wbgapi` package)
- UN data downloadable from data.un.org
- Census data available from national statistical offices' websites

Dependent variables (Urbanization Measures):

1. Urbanization rate (%) = (Urban Population/ Total population) × 100

2. Urban growth rate (%) = $\frac{[(\text{New_Urban_Population} - \text{Old_Urban_Population})/\text{New_Population}]}{1} \times 100$
3. Urban Primacy Index = $\frac{(\text{Largest city Population}/\text{Total Urban Population})}{1} \times 100$
4. City Size Distribution = Number of cities by population category which ranges to
 - i. Less than 100,000
 - ii. From 100,000 to 500,000
 - iii. From 500,000 to 1,000,000
 - iv. From 1,000,000 to 5,000,000
 - v. Above 5,000,000

There are several independent variable such as

1. Economic indicator
 - GDP per capita
 - Sectorial employment distribution
 - Poverty headcount ratio
 - Foreign Direct investment
2. Demographic indicators
 - Total fertility Rate (TFR) births per women
 - Life Expectancy at birth (years)
 - Median age
 - Natural increase rate (births – death per 1000)
3. Infrastructures indicators
 - Access to electricity (% of urban center to total population)
 - Access to improved water (% of urban center to total population)
 - Mobile phone subscriptions (per 100 people)
 - Road density (kilometer per 100 square kilometer)

DATA DESCRIPTION

The differential population growth across Nigerian states shows that Lagos and FCT experienced the most rapid urban population expansion between 2000 and 2020, with urban populations growing at compound annual growth rates exceeding 3.5%. Furthermore, the population growth Trajectory shows how fast each state's population has grown over the 20-year period with Urban-Total Gap (vertical distance between the two lines) represents the rural population. The urbanization Pace in each states where the urban line closely follows the total line are highly urbanized showing the comparative scale which allows easy comparison of population sizes across states

EXPECTED RESULT

Lagos state exhibits the highest absolute urban population growth, expanding from about 6.4 million in (2000) to 13.0 million (2020), showing a compound annual growth rate of about 3.6%

(UN, 2024) with the federal capital territory, Abuja showing the highest urbanization rate of about 96% in 2020, a direct impact of its position as Nigeria administrative capital with limited rural area (Bloch, R., et al., 2015). In Rivers state, it is expected to display a slightly balanced urbanization level of about 71% driven by Port Harcourt oil-driven economy. Conversely, Oyo state is has the lowest urbanisation rate of approximately 53%, reflecting a more balanced population distribution compare to Port Harcourt. Oyo state urban-rural population is engineered by it traditional agricultural economy (Gandy, 2006). In addition all the four states demonstrated a consistent urban growth from 2000 to 2020 with little to no observed reversal, aligning with the National Population Commission.

Total vs. Urban Population Growth Patterns

Divergent growth trajectories across the four selected states is shown in chart 1 (as provided from the python code). Lagos is primarily an urban character indicating the fact that it will exhibit the steepest rate of urban population growth as well as the narrowest gap between the total and urban population line (Jiboye, 2011). All through the period, the FCT's line was almost overlap and from 2000, it was observed that more than 90% of the population was living in the urban areas. Oyo state in the other hand continues to have the largest percentage of rural residents over the study period. All the states will see a steady increase in the shaded areas that represent total urban population, with Lagos and the FCT exhibiting the most noticeable growth.

Urban Primacy and Population Distribution

Unlike the analyses on the national level, state-level will not assess urban primacy using conventional metrics. Chart 10 shows the ranking over the period of time, Lagos maintained the population increase from about 6.4 million to over 13 million, Rivers state, Oyo and the Federal Capital Territory (FCT) increased from about 2.3 million to 5.2 million, 2.0 million to 4.2 million and 0.7 million to 4.2 million respectively according to National Bureau of Statistics, 2021. With the fact that the FCT has the highest urbanization rate, it is ranked fourth in the absolute population size as a result of significantly smaller overall population. This overall undermines the crucial difference between urbanization rate, which are expressed in percentages and not in absolute urban population size.

Urban Growth Rates

Has shown in chart 5, it was demonstrated that urban population growth continuously outpace the overall population growth in all the states. Urbanization is driven rural to urban migration rather than natural increase (Harris, J.R., et al 1970). As a result of capital rapid development phase from 2000 to 2015, FCT has the highest and most variable urban growth rate (10% to 15% in a five-year period). Contrarily Rivers and Oyo exhibit more balanced rate of (3% to 5% annually), while Lagos continues to exhibit high urban growth rate of approximately 4% to 5%.

Urbanization Rate Trajectories

Four different urbanization paths were visualized in chart 2, the study's most important visualization. FCT has continued to hold the top spot of about 88% to 96% and its comparatively flat trajectory implies that it was already heavily urbanized by 2000. Lagos shows a steadily rising trend from 80% to 90% crossing over the 85% threshold around 2010. In all of the four states, Rivers exhibits the fastest urbanization transition (+14 percentage point), with a moderate upward trajectory from 57% to 71%. From 45% to 53%, Oyo shows the least significant increase, maintaining the closest trajectory to Nigeria's national average of 52% (UN, 2024). These states are more urbanized than Nigeria's ruraly-dominated states.

Urban-Rural Compositional Changes

As shown in chart 6 (stacked area composition), the urban expansion accounted for more than all of Lagos population growth from 8.0 million to 14.4 million, with the quantity of green (rural) space leaving the small and essentially part of approximately between 1.4 million to 1.6 million. The most noticeable change was seen in the FCT, where the green areas were almost disappearing while the urban centers grow quickly. Although the urban growth is more than the rural growth rate, Rivers is showing a balanced growth in both components. Oyo, boasting as an agricultural hub among the four states and an urbanizing state, continues to show significant growth in both urban and rural population, with green space (rural) growing concurrently with the blue space (urban) (Owusu, 2005).

Absolute Change in Urbanization

According to chart 9, Rivers state has undergone the largest urbanization transition (+14 percentage point, rising from 57% to 71%), closely followed by Lagos (+10 percentage point from 80% to 90%), FCT and Oyo show a moderate of (+8% points each). Given the high baseline of 88% in 2000, FCT's slight changes suggest that there is little room for further growth that is called "the ceiling effect" associated with densely populated areas. In line with its traditional economy and robust rural sector, Oyo's smaller change suggests slower urbanization dynamics (Cobbinah, P.B., & Amoako, C., 2012).

Statistical Analysis Results

In order to verify that the urbanization rates in the four states are actually different urbanization models rather than random variation, the statistical analysis section's ANOVA test should reveal statistically significant differences ($p < 0.05$). According to (Henderson, 2003), correlation analysis is likely to show a strong positive relationship ($r > 0.6$) between the total population size and the rate of urbanization for Lagos and the Federal Capital Territory, but weaker correlations for Rivers and Oyo. This suggests that factors other than population size, such as administrative functions and economic structure, are what drive urbanization. According to the decade-level urbanization growth analysis, all four states saw steady increases in urbanization, with Oyo and FCT each seeing about 4 percentage points per decade, Lagos 5 percentage points per decade (steady transformation), and Rivers averaging about 7 percentage

points per decade (fastest transition). These rates are higher than Nigeria's national decade average, which is roughly 3.5 percentage points, indicating that these states are leading the nation in urbanization.

Comparative Rankings and Dynamics

The states will be ranked clearly in Chart 4 (2020 urbanization rate comparison): The most and least urbanized areas are separated by 43 percentage points, with FCT (96.0%) > Lagos (90.2%) > Rivers (71.0%) > Oyo (53.0%). This hierarchy illustrates how urbanization is primarily driven by administrative (FCT) and commercial (Lagos) functions rather than industrial (Rivers) and traditional agricultural-commercial (Oyo) economies. Chart 3 (urban vs. rural comparison) will show Lagos housing approximately 13M urban versus 1.4M rural residents (9:1 ratio), the most extreme urban dominance, while Oyo will exhibit the most balanced distribution (approximately 4.2M urban vs. 3.7M rural, 1.1:1 ratio).

Best Practices and Urbanization Models

Every state has a unique model with equally unique benefits and challenges, no state can stand out as having the most “sustainable” urbanization trajectory based on the results. Lagos successfully absorbs large population flows, but it also encounters congestion and strain on its infrastructure (Gandy, 2006). With limited rural economic integration, FCT achieves the highest urbanization efficiency with planned development. With the volatility caused by oil dependence, Rivers strike a balance between rural retentions and urban growth, also oyo show a resilience bur slow economic transformation as a result of its strong agricultural base and blue to green ties. Rivers states is a resource driven urbanization while Lagos teaches scaled management, FCT teaches planned development and Oyo state teaches balanced regional development (Jiboye, 2011).

Policy Implications

The result suggest a differentia policy is to be used. Lagos and FCT with over 90% urbanization rate requires a mega city management approach with include quick transit network, vertically erected housing development, good waste management system and metropolitan governance reforms. Rivers state with over 70% to less than 80% urbanization rate, a need for secondary city to strengthen areas surrounding Port Harcourt, which is known for its industries of various diversification outside of oil, also the peri-urban planning needs proper expansion management. Oyo with less than 60% urban growth rate, needs to improve on the rural-urban linkage, small town development and agricultural value chain enhancement as well as preventing excessive primacy of Ibadan (Bloch et al., 2015)

Abuja, Lagos, Oyo and Rivers needs to address the fundamental issue of informal settlement expansion, with Lagos and Rivers experiencing the most acute challenge given their rapid growth rate. It is observed that the urban growth rate exceed the total population growth rate across all states confirming that rural-urban migration remains the primary source of urbanization, making

the need for policies that will create and explore more opportunities in rural area rather than trying to manage the urban influx. The comparative analysis across these four states show a microcosm of Nigeria broader urbanization challenges with lessons for the other state of the nation.

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(archived data)